



Keeping the heart a long way from the brain: The emotional labour of climate scientists



Lesley Head ^{a, *}, Theresa Harada ^b

^a School of Geography, University of Melbourne, 221 Bouverie St, Carlton 3053, Australia

^b Australian Centre for Cultural Environmental Research (AUSCCER), University of Wollongong, Wollongong 2522, Australia

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ABSTRACT

This paper contributes to increased understanding of emotions and climate change through a study of the emotional management strategies employed by a sample of Australian climate scientists. We bring three broad areas of literature into conversation in order to think more productively about climate change and emotion: recent applications of the concept of emotional labour, studies of the role of emotion in science, and feminist perspectives on the performative role of emotions. In response to contextual drivers that include the social norms of science, a strong climate denialist influence and the preservation of self and family, these scientists mobilize a range of behaviours and strategies to manage their emotions around climate change and the future. These include emphasizing dispassion, suppressing painful emotions, using humour and switching off from work. Emotional denial or suppression of the consequences of climate change worked to enable the scientists to persevere in their work. This study suggests that painful emotions (anxiety, fear, loss) around climate change need to be acknowledged and discussed.

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1. Introduction

There is consensus among the majority of scientists that anthropogenic climate change is already leading to increased temperatures and the 'likelihood of severe, pervasive and irreversible impacts for people and ecosystems' (IPCC, 2014, p. 8). Despite this, there is little evidence of effective political or social change; it is very much 'business as usual'. In order to avoid two degrees of warming over pre-industrial levels requires that a third of known reserves of global oil, half the gas and over 80% of coal remain unused (McGlade and Ekins, 2015). Yet many scientists think it may already be too late to avoid two degrees of warming due to the lag effects of emissions already in the atmosphere (Anderson and Bows, 2011), which means we are on track for four to six degrees of warming with an increase in extreme events, and fundamental changes to society and its ecological underpinnings. It would be surprising if these predictions did *not* generate emotional responses, but the ways in which emotions are discussed in relation to climate change are quite circumscribed. In particular, emotions

are commonly separated from the construction of climate change as a scientific, rational issue. Media depictions of climate change rely on science as the ultimate knowledge source, legitimizing predictions and reinforcing the social power of scientific certainty (Carvalho, 2007; Cass and Walker, 2009). The expectation from policy-makers and the public is that the engagement of science with environmental and other issues will not be emotional (Cass and Walker, 2009). Yet only relatively recently have there been calls for greater attention to the emotional dimensions of climate change and climate science. These calls suggest relationships between these themes and the management of mental health and well-being in daily life (Farbotko and McGregor, 2010; Consulo Willox, 2012).

An emerging literature identifies a number of emotional dimensions to climate change: emotions as a source of denial or blocking of information (Norgaard, 2006, 2011), widespread feelings of anxiety and loss (Randall, 2009; Berry et al., 2010; Doherty and Clayton, 2011), mourning for lost places or relationships with the natural world (Furberg et al., 2011; Adger et al., 2013; Drew, 2013), recovery from the impacts of climate events (Convery et al., 2008; Mort et al., 2005; Tapsell and Tunstall, 2008) and a generalised grief about the future. Researchers are increasingly prepared to discuss their own emotional responses to traumatic

* Corresponding author.

E-mail addresses: lesley.head@unimelb.edu.au (L. Head), Tmh952@uowmail.edu.au (T. Harada).

subject matter related to climate change and associated environmental disasters (Willis, 2012; Dominey-Howes, 2015). Nonetheless, painful or troubling emotions receive little attention in the western cultural context of the practice of science, which is dominated by the importance of scientific rationality. When considering the uncertainty of environmental futures the focus has tended to be on ideas that emphasize resilience, adaptation and risk management, which one could argue are associated with a positive emotional outlook (Rickards et al., 2014).

In this paper we seek to contribute to increased understanding of emotions and climate change through a study of the emotional management strategies employed by climate scientists. Climate scientists provide an important empirical study because they experience the intensity of the issues and debates on a day to day basis, in a rational scientific context which requires them to downplay their emotions. Some key questions for us are, what kind of emotional management is being undertaken by those at the heart of the rational endeavour (Whittle et al., 2012)? What kinds of feeling rules are they following (Hochschild, 1979), and what are the tools in their toolkit of emotion management (Norgaard, 2006)? Before proceeding to the empirical study we outline three broad areas of literature that the paper brings together; recent applications of the concept of emotional labour to climate change, studies of the role of emotion in science, and feminist perspectives on the performative role of emotions.¹

1.1. Emotional labour and climate change

The concept of emotional labour was originally developed by Hochschild (1979, 1983) to focus on the processes by which people manage their emotions in different contexts or situations in order to conform to what is deemed socially appropriate. Initially, the work of managing emotions was viewed differentially through a demarcation of the private and the public sphere and named as either work or labour. A diverse range of literature has examined the emotional management of self and others, and the ways this can serve the interests of, and be used by, other parties, for example the capitalist commodification of emotions in the service industry (Blackmore, 1996; Fineman, 2000; Mann and Cowburn, 2005). Whittle et al. (2012) argue that:

Emotion work can therefore be internal or external, and involve either evocation (trying to induce a desired feeling) or suppression (trying to quash a feeling that we consider to be undesirable). (Whittle et al., 2012: 62)

While some literatures distinguish between the concepts of emotional work and emotional labour, depending on whether a domestic or professional context respectively is being discussed, we prefer to consider emotional management as a set of broader coping strategies that incorporate both emotional work and emotional labour at different times and to different degrees. We agree with Whittle et al. (2012) that the boundaries between work and home, or physical and emotional recovery, are often blurred and that there is much crossover in how, where and by whom emotional management is deployed.

As Hochschild's original paper discussed, there can also be varying levels of authenticity in the performance of emotions, and later critique and counter-critique (Brook, 2009) has increased the nuance with which the performativity is analyzed (Bosco and Joassart Marcelli, 2015). As such, in this paper we also consider how emotional management strategies may be aligned with authenticity, that is how climate scientists view their underlying personalities in relation to the performance of emotion as they deal with the stresses of climate change knowledge.

One of the most important studies of emotional management in relation to climate change has been Norgaard's (2006, 2011) ethnography of the pseudonymous Norwegian town of Bygdaby. She argued that climate change denial was not a matter of individual psychology but was socially organized and reproduced.

To say that there were unpleasant emotions associated with global warming is not enough to explain the lack of social movement activity in Bygdaby— especially as such emotions can also serve as the impetus for social action (Norgaard, 2011, p. 94).

Norgaard recognized that Norwegians were in a dilemma; they did not dispute the validity of climate change science yet their economy had benefitted significantly from North Sea oil reserves. Norgaard coined the term 'double reality' to describe the social process of everyday denial that took place through norms of conversation and norms of emotion – the 'tools' in their toolkit of emotional management. In response to the emotions of fear and helplessness, the norm for townspeople was to be optimistic and maintain control. They employed 'selective attention' as an emotional management strategy (controlling exposure to information, not thinking too far ahead, focusing on something you can do). When dealing with guilt, townspeople employed 'perspectival selectivity' to reinforce national pride (we are not as bad as the Americans, and Norway is a little country anyway) (Norgaard, 2006, Table 4). The understanding of denial as selective attention has also been put forward by Opatow and Weiss (2000, p. 479); 'denial is a form of selective inattention toward threat-provoking aspects of a situation to protect a person from anxiety, guilt, or other ego threats'. It is clear that when dealing with the emotions associated with climate change that individual management strategies are embedded in wider social understandings of how to appropriately respond.

1.2. Emotions, climate change and science

At the political level a discourse of fear and risk has been influential in framing global climate change (Beck, 1999, 2002; Hartmann et al., 2005; Pain, 2009), even in the face of a lack of political action. Yet some have argued that 'fear won't do it' (O'Neill and Nicholson-Cole, 2009), and that emotions such as fear, anxiety and grief are paralyzing rather than productive (Roser-Renouf and Maibach, 2010). Nevertheless the documentation of emotions around loved places (Adger et al., 2013; Barnett and Campbell, 2010; Drew, 2013; Hastrup, 2009; Moser, 2013) or degraded and diminished natural environments (Brown and Pickerall, 2009; Consulo Willox and Harper, 2013; Wiseman et al., 2010) are important aspects of this literature. These studies illustrate the importance of emotion in how people negotiate everyday spaces and social processes which are ultimately situated within wider policy arenas (Anderson and Smith, 2001; Davidson and Milligan, 2004; Parr et al., 2005; Smith and Pain, 2008).

When we bring this evidence into conversation with the dispassionate norms of science we can better understand the influential role of climate scientists in framing our understanding of

¹ This topic is of broad interest. A number of journalists have interviewed climate scientists about their emotional responses to the future, as in this example titled 'When the end of human civilization is your day job' (Richardson 2015). At the Melbourne Writers Festival in August 2015 an exhibition of 22 handwritten letters from some of Australia's leading climate researchers describes how climate change makes them feel (Melbourne Writers Festival, 2015). A similar website (<http://www.isthishowyoufeel.com/this-is-how-scientists-feel.html>) includes Australian climate scientists among an international group.

the climate change challenge. The importance of the dispassionate norm in science is emphasised by [Brysse et al. \(2013\)](#), who demonstrate how it leads to scientists being 'biased not toward alarmism but rather the reverse' (p. 327). Using examples from predictions of Arctic ozone depletion and the possible disintegration of the West Antarctic ice sheet, they argue that

the scientific values of rationality, dispassion, and self-restraint tend to lead scientists to demand greater levels of evidence in support of surprising, dramatic, or alarming conclusions than in support of conclusions that are less surprising, less alarming, or more consistent with the scientific status quo. Restraint is a community norm in science, and it tends to lead many scientists ... to be cautious rather than alarmist, dispassionate rather than emotional, understated rather than overstated, restrained rather than excessive, and above all, moderate rather than dramatic. ([Brysse et al., 2013](#): 327–8)

In practice, where social studies of science have turned their attention to the importance of emotions in the doing of scientific work ([Barbalet, 2011](#); [Shapin, 2010](#)), they have found emotions to be pervasive in science. [Spencer and Walby \(2013\)](#) elaborated three specific forms of emotion. These included those related to how the community shared success and failure, overcame frictions and obstacles within and between groups, and negotiated non-work demands. Emotions were acknowledged at multiple levels as important to how scientists negotiated the boundaries between work, home and wider social and political arenas.

To make sense of a context where reason and rationality generate data that can give rise to significant emotions, including painful ones, it is important to find a framing that can bring these two things together rather than see them as contradictory. Much of this has been due to feminist approaches, to which we now turn.

1.3. Feminist approaches to emotion and reason

Feminists have long argued that the emotional is missing in many accounts of the socio-spatial world ([Anderson and Smith, 2001](#); [Davidson, Bondi and Smith, 2005](#); [Turner and Stets, 2005](#)). Feminist analyses of emotion have shown how it is viewed as separate from and

'beneath' the faculties of thought and reason. To be emotional is to have one's judgement affected: it is to be reactive rather than active, dependent rather than autonomous. ([Ahmed, 2004a](#): 3)

Emotions have been understood as primitive and corresponding to the animal-like stage of evolution and often as the antithesis of rationality ([Ahmed, 2004a](#): 3). This binary of the rational and the emotional does little to advance understandings of how people make sense of themselves and their place in the world ([Smith and Pain, 2008](#)). A feminist attention to emotion can uncover how this may work to orient people towards objects, communities, practices or ideas, and mediate how they view themselves as members of particular families, groups, communities and nations ([Bondi, 2005](#); [Conradson and McKay, 2007](#); [Thien, 2005](#)).

Importantly in these analyses, to be emotional does not imply irrationality. At times rationality can contain emotion, as in how people might choose a strategy of denial to deal with fear and anxiety. For example [Koskela](#), in the context of women's personal safety, argued 'Reasoning can be thought of as a strategy to contain fear, but also as a way of maintaining sufficient courage to avoid fear and, thus, to live without it' ([Koskela, 1997](#): 306). What is significant here is how people deal with emotion in rational ways,

making practical choices about the best way to deal with experiences of fear, anxiety, love or pleasure. As [McKinnon \(2014: 31\)](#) says, 'It is not obvious that people in despair about tackling climate change are making a mistake.' Rather, we suggest that they are taking a rational approach to managing their emotional responses. It is this emotional work that we wish to map.

We employ [Ahmed's](#) conceptualization of emotion as something that emerges in 'in-between' spaces that is, between subjects and between subjects and society. In this way of thinking, the subject and the social cannot be considered in isolation. Rather we can consider the way that emotions work in relation to how we understand the boundaries between the public and the private, and between individual bodies and particular 'collectivities', as this has significance for thinking about climate change. Emotions here are performative in that they work to define the boundaries of subjects and orient people towards particular collectivities.

In the case of climate scientists we can understand how emotions or 'intensification of feeling' can work to define and maintain boundaries- that is to produce subjects (the 'good' scientist') and objects ('hard' data) and collectivities (responsible and reliable climate change advocates). A key point we are making in relation to how we approach the discourses of climate change is that more attention should be paid to the way that climate change materializes our fear, anxiety, grief or resentment. An important challenge is to more carefully map the performative role of emotion across bodies and collectivities.

2. Methods

The work with climate scientists was part of a broader project examining hope for the future and emotions surrounding climate change. Climate scientists were thought to be a particularly interesting study because their daily work is to amass the evidence with implications that challenge the rest of us. An important part of much recent scientific work in climate change has been to communicate it in summary terms for policy-makers and the broader community. We approached a sample of Australia's leading climate scientists, based on the most prominent names in the published literature. The thirteen who agreed to be interviewed (four female and nine male) constitute roughly 50% of the nation's leaders in this field. Ages ranged from the early career stage (defined in Australia as within five years of PhD completion) to retired but still academically active. At the time of interview they were employed in, or had adjunct appointments at, universities and/or government scientific bodies. The term 'climate scientists' as used in this paper encompasses two main areas of expertise. Some interviewees were atmospheric climate scientists with disciplinary backgrounds in physics and maths. Others had more general backgrounds in the earth, environmental and biological sciences, and tended to work more on the implications of climate change.

Interviews took place in or close to participant workplaces in eastern Australia between June and October 2014. Core questions included how their thinking around climate change had evolved over the course of their career, how they saw themselves on a continuum between optimism and pessimism, how they saw the future, what kinds of social and policy changes were necessary to deal with climate change, and how likely these were. Follow up questions responded to specifics of the individual answers. (Note that this was not a study of how science is done, as undertaken for example in Actor-Network-Theory ([Latour, 2005](#))). Interview transcripts were coded independently by the two authors, and results compared. Themes that structure the results section below are a combination of those framed by the questions, and those that emerged in the conversations. In the quotations used below, pseudonyms are used for individual responses that exemplify

wider trends. (In order to preserve anonymity within this small community, more detailed demographic descriptors are not used.)

3. Results - context and strategies

We identified several common explanations driving emotion management, and identified a range of emotional management strategies (see Table 1). Common causative triggers included: the social norms of science, the weight of denialist discourses and the obligation to protect self and family. We discuss each in turn and provide examples of the scientists' reasoning and behaviours.

3.1. The social norms of science: 'keeping the heart a long way from the brain'

One of the most common themes to emerge was that rationality was the key to good science. In a variety of ways all the interviewees spoke of the crucial importance of doing rigorous science as underpinning everything their work entailed, and the basis of their professional reputations. Geoff described this as 'keeping the heart a long way from the brain'. Here he complies with the scientific norms of the doing of climate science – that science is based on observable facts about the physical world and the inherent properties of things – which requires an objective standpoint (Mulkay, 2015). For Geoff, emotions were to be distanced from rational thinking in order to produce accurate reliable data. In a similar way, Garry acknowledged the emotional dimensions of the scientific projections and data that he helped to produce, yet illustrated how he finds it necessary to actively eliminate these, saying: 'I would prefer to describe what constitutes a catastrophe rather dispassionately, without taking on the emotional impact of that.'

Also conforming to the social norms of science, Ed spoke of the possibility of being wrong as the only thing that kept him awake at night. Thus it was not fear and anxiety about the possible consequences of climate change that troubled him most, rather it was the accuracy and reliability of his calculations and modelling that were at the heart of his anxiety. Here we can understand how the professional objectivity required to ensure reliability and accuracy of data can be considered as part of the emotional labour for this group of climate scientists.

3.2. Denialist discourses: graveyard humour

Second, there was a shared understanding among the sample that the strong climate change denialist movement was a source of pressure and a cause of anxiety:

Prominent people are accusing you of being a fraud ... that's going to weigh on you ... And we used to get lots of hate mail and death threats; that seems to have stopped a year or two ago ... it's not something that other scientists have to deal with. (Brian)

Dealing with 'hate mail' and 'death threats' and the onus to 'defend' their science was a common theme and a source of resentment for many. Geoff illustrates this by saying:

Almost all of the fundamental science that gives us clear evidence for global warming doesn't come from climate science. It comes from chemistry, physics, mathematics, statistics and yet we're held to account for failing to be able to articulately explain the relationship between quantum theory, the rotation of a CO₂ molecule and its ability to absorb infrared radiation ... The physicists do not have to deal with deep scepticism on whether the universe is expanding. The aeronautical engineers do not have to defend whether the shape of a wing generates uplift. (Geoff)

As Geoff makes clear, climate scientists are being held to public account in ways that other scientists are not. One blamed the 'relentless attacks' from denialists, the media and politicians for creating a negative atmosphere in the whole area of work:

you can't have kept on doing this for a long time without having a bit of ... graveyard humour ... If you let it get you down then you're gone, you're no longer with us. The ones that have remained are the ones who are sufficiently stoic about this to laugh off the regular attacks ... If that sort of thing gets you down then you're no longer a climate scientist, you've moved onto something less continuously negative I guess. (Martin)

We can see that the tactics for dealing with the negativity of denialist discourses include developing a sense of stoicism, being able to laugh it off with 'graveyard humour' and developing a 'thick skin'. These strategies help the scientists to deal with the anxieties and stresses that arise from their immersion in a denialist context; it is through emotional management that they maintain their capacity to work productively as climate scientists. Australia is one of the western countries where consistently high levels of climate change denial have been recorded (Tranter and Booth, 2015), and there has been an active denialist movement (Oreskes and Conway, 2012). What is interesting here is how this context further entrenches a downplaying of public emotion around the severity of the problem while highlighting the challenge of dealing with personal emotional responses.

Table 1
Common themes from interview data, and the reasons for and strategies of emotion management.

Causative theme	Explanation and/or reason	Response and/or strategy
Social norms of science	Dispassion and rationality understood as central to the role of the scientist in producing reliable data.	Blocking or not acknowledging emotional responses to confronting data about the consequences of inaction on climate change.
Denialist discourses	Discourses of the uncertainty of climate change predictions were seen to undermine the legitimacy of scientific authority. This was seen as particular to climate change science rather than other sciences (e.g. physics, maths, chemistry).	a) Developing a 'thick skin' to criticism or threats- ignoring media criticism. b) Avoiding public advocacy for fear of retaliation.
Maintaining work life boundaries	Scientists generally partitioned work and home, though some acknowledged the difficulty of not thinking about work.	Engaging in everyday banal activities (e.g. reading detective novels, gardening) to reduce stress.
Protecting the psyche	As a part of self-care scientists tried not to think about the consequences of climate change.	Blocking out thinking about climate change even when confronted with evidence outside the work environment (e.g. changing conditions in the local beach environment).
Being a good parent	Refraining from discussing the implications of climate change.	Scientists routinely chose not to discuss climate change with their children in order to shield them from anxiety and fear.
Self-preservation	Scientists recognized their privileged positions (Western, well-educated, generally wealthy).	Formulating strategies for escaping the consequences of climate change (dual passports, housing in alternative locations).

3.3. *Protecting self and family: I can't do it 100% of the time*

Third, under the theme of preservation of the self and family, participants described quite emphatically the emotional dimensions of climate science outside the workplace and the range of strategies that were needed to cope with this. Many avoided discussing climate change in social situations, when they would 'rather talk about the cricket' (Nick). Karen illustrates how social situations have the potential to become a site of conflict and anxiety due to the nature of her work.

Oh God I never tell them [in a social situation, that I'm a climate scientist], it usually descends into them shouting at you, or yeah, like telling you either you're not doing enough or you're doing too much. (Karen)

This quotation illustrates how Karen aligns herself with the scientific community as opposed to the 'them' of the broader community. It suggests that embodied difference comes to be felt in the meeting with what is deemed to be 'outside' or 'the other' in how she understands herself as a climate scientist. It is an example of how emotion works to 'define the contours of the multiple worlds that are inhabited by different subjects (Ahmed, 2004b: 25).' Scientists employed different strategies to avoid discussing their work, retreating from their expertise and embracing the banal. They adopted a non-committal stance in social situations. For example Susan said, 'I don't talk to taxi drivers ... I love a silent taxi driver'.

Despite this, most of the scientists described always being 'on' and not able to easily separate these work and home subject positions. Like many other hard working professionals, attempting to compartmentalize work and home was seen as a strategy for self-preservation.

I go home, that's home. It's not work ... I want to read trashy novels and have cups of tea. (Karen)

I actually deliberately block it out as much as I can [when away from work]. So I try, I really try not to because I don't think I'd be in very good shape if I let myself think about it all the time. So there's a denial for you [laughing] ... I have to say to be brutally honest I tend not to look at [the future] too much because it's very confronting. I have kids ... (Linda)

Hochschild (1979) argued that when we inspect and evaluate our emotional state from an objective point of view we are using a 'feeling rule' as a part of emotional labour. Linda and Karen both illustrate this cognitive processing of emotions as a strategy to forestall the negative impact of contemplating climate change in the light of their scientific knowledge. The 'feeling rule' is not to allow negative emotions to permeate the personal and private, but to embrace the routine and unexceptional of the everyday to sustain a sense of self.

In the light of these strategies, protecting the psyche from the subject matter of climate change was no easy task. It created a 'dissonance' from the evidence at hand because of the severity of the implications and called for a rational approach to emotion. For example, Susan compared it to the way we think about death:

we all know we're going to die but most of us who are reasonably socially well-adjusted don't think about it every day because it's paralyzing, and denialism is a form of protecting people against paralysis I think. (Susan)

Similarly to the townspeople of Bygdaby, scientists used a form of personal denial to protect the psyche (Norgaard, 2011). This was seen as a necessary strategy to manage emotions both as an individual, and as a member of an influential group. For this group of scientists, it was necessary to collectively suppress the emotional

responses of fear and anxiety evoked by the data themselves in order to continue to produce reliable scientific data. Recognizing the emotional dimensions of climate change led scientists to personally manage emotions with specific tools which ultimately strengthened group resolve and perseverance.

Further to preservation of the self, emotional management strategies were extended to include other family members, especially children. The reticence to discuss climate change issues with children arose a number of times, and the parents who mentioned this were worried about leaving children and teenagers distressed and disempowered. Thus scientists might be seen to be suppressing their own emotions in order to shield their children from anxiety and this is a part of their performance of being a responsible parent. As Opatow and Weiss (2000) argued, it is important to acknowledge and identify the diverse manifestations of denial in environmental conflicts in order to be able to manage them. What we see in the scientists' refusal to discuss climate change with children is the invisible work of emotional management (Whittle et al., 2012) and a kind of 'moral exclusion' where some people are deemed to matter more than others (Opatow and Weiss, 2000).

3.4. *Optimism and pessimism in relation to scientific knowledge*

We asked scientists to place themselves on a continuum between optimism and pessimism in their feelings about climate change, in order to gauge their level of hopefulness about the future in relation to their expert knowledge of probable future outcomes. Psychological definitions of optimism as a disposition are associated with motivation and confidence that a goal can be attained and imply perseverance even in the face of adversity, whereas pessimism implies doubt that the goal is attainable and a gradual or total withdrawal of effort toward achieving it (Carver et al., 2010). We thought it important to map out how confident this group of scientists were, given their expert knowledge, that a solution to the challenges of climate change could be found. We found a broad spread of answers including 'it depends on the day', and descriptions of feelings as 'wildly oscillatory', 'schizophrenic' and 'mixed'. No one was unambiguously optimistic, and there were many more expressions of pessimistic than optimistic thought. Some claimed that their pessimism had increased over the last few years due to inaction by government and policy makers, indicating 'a gradual move towards pessimism'. Alongside this there was widespread frustration that people, especially governments, were not 'taking notice' of the science. We might expect such pessimistic dispositions to demotivate scientists in their work but what we found was the reverse. For example, despite high levels of expressed pessimism Ed describes how he actively manages this in order to stay motivated:

So you kind of have to be optimistic because if you fall into at least acting from a state of pessimism then you can't break out, you get into a downward spiral. It becomes a sort of, I guess at first a personal weight, but also a weight that gets transmitted from one person to another and therefore, stops us from doing rational things on climate change. And to the extent that we need to do rational things on climate change, it's really important that we're not depressed collectively about the future. That if the future becomes something that's too big to handle and outside our control then we've lost it and so, that's why I sometimes say I'm a compulsory optimist. (Ed)

Ed demonstrates how emotional management helps to curb a disposition towards pessimism; he understands a rational stance is needed to maintain personal and group resolve. The performance of an optimistic disposition works to counter pessimism, which as Ed comments 'gets transmitted from one person to another'.

Similarly, Susan argues:

once you become convinced that the world, the way it's going, is doomed, the human race is doomed, I don't see how you can do anything other than keep trying and work in this area. I don't understand why all scientists aren't working on climate change. I honestly don't, because if you don't have a habitable planet, it doesn't matter ... nothing else matters ... We really are screwed if we all give up. (Susan)

For Susan and Ed, emotional management strategies helped to reinforce group strength and resolve – by overcoming a pessimistic outlook their compulsion to continue to work was intensified.

3.5. Pleasurable emotions

We want to stress that not all emotional responses to climate change expressed by the scientists were negative. There were many expressions of the pleasurable emotions that were associated with their jobs and interviews were often interspersed with laughter and humour. On the whole scientists expressed pleasure when describing their work and spoke about science as being 'cool', 'interesting' and fun:

I got involved primarily in a science perspective on climate change because I felt that it was a really interesting science problem. (Thomas)

... the science is so intriguing, so interesting because it's such a vital issue and it's ... fascinating ... you can get really fascinated with how this planet's changing and how we're pushing it. (Nick)

These pleasurable emotions were undoubtedly another way that the collectivity of the scientific community was reinforced. The emotions associated with the challenges of climate science helped to inspire and motivate scientists to 'keep on going'. Rationally they understood there were few alternatives:

We've just got to get on with it ... The government could be helping a lot more, but actually we've got to get on with it anyway. (Paul)

We really are screwed if we all give up. (Susan)

Ahmed argued that '[E]motions, then, are bound up with how we inhabit the world 'with' others (2004b: 28).' Turning away from an object of fear (here the consequences of climate change) can evoke 'fellow feelings' and a reiteration of signs of 'fellowship' which may also be understood as compulsory, so that for scientists, this fortitude to persevere was one of the ways that emotions worked to bind them together. Emotional management was crucial to all of the interviewees in a variety of ways, and while we could say this of the rest of society, what was significant for this group is that they confront the climate change data on an everyday basis.

4. Discussion

Notwithstanding both the demise of the mind/body dichotomy in biology, and increasing analysis of the role of emotion in scientific research, these scientists work in a context where the predominant view denies or hides the emotional dimensions of doing science. In order to maintain their credibility the scientists have to implement a number of emotional management tools in the labour of dispassion. This emotional norm is influenced by complex factors that include (i) a view of science in which the heart is kept a long way from the brain, to paraphrase one participant, and (ii) a

broader social context in which climate denialism is strong and scientists are still having to debate the 'truths' of anthropogenic climate change. Our analysis does not permit us to disentangle the relative contributions of these two main factors, but it is important to note that they both work in the same direction, which is to make it easy to dismiss displays of emotion as irrational and irrelevant. No doubt this effect is exacerbated by a broader masculinist context in both scientific and denialist groups. While emotional labour is argued to be very gendered (James, 1989), we have focused here not on any gender differences but on the norms against which the scientists discipline themselves.

The scientists' aspirations to separate themselves from emotions are demonstrated by their diverse distancing practices. These include not only upholding the identity and reputation of the objective scientist, but also the need to maintain personal well-being in demanding professional lives, and a protection of the psyche against worst case scenarios that some participants referred to as denial. This distancing is a complex process; while the expectation of themselves and others is that they privilege rationality and objectivity, they were not immune to a range of emotions in their everyday work lives, including anger and frustration that the rest of society is not taking sufficient notice of their knowledge. The multiple ways that the scientists distance themselves from emotions around their work can be understood as constituting a kind of everyday denial, as conceptualized by Norgaard (2011). There was resonance here in the reluctance in both studies to discuss climate change with children.

The labour of optimism and its companion, the suppression of doom and gloom, conform to feeling rules frequently observed in western culture; the expectation to manage emotions to remain optimistic and to thus emphasize pleasurable emotions over painful ones. This broader cultural dispossession contributes to downplaying potentially severe consequences, consistent with Brysse et al.'s (2013) argument for 'Erring on the Side of Least Drama' (ESLD). In the context of the inherent conservatism of the scientific method, several aspects of our findings resonate here, including the distancing from catastrophist thought ('doom and gloom') and the 'compulsory optimism' of keeping going regardless. As another example of the pressure to lean towards the positive, participants frequently referred to social tipping points in an optimistic way, mainly in relation to the uptake of solar PV. But few referred to the possibility of tipping points in the opposite direction of social catastrophe and breakdown (beyond generalised statements like 'it won't be pretty' or the future is 'bleak').

On the other hand, emotions expressed as 'love' or 'fascination' for their work also worked to stimulate engagement and connection; being absorbed in and captured by the nature of the work was a positive aspect of being a climate scientist. It is important to emphasize here that the sample is already skewed towards personal resilience and the ability to survive and succeed in a competitive academic environment of limited resources. Anyone working in a university science context would recognize this male-dominated sample. The number of opportunities shrink at each career stage, creating a funneling effect, with even the early career participants here already survivors of such a system. This perhaps reinforces how emotional management strategies have contributed to attaining their current positions.

These climate scientists distance themselves from stress and anxiety by downplaying the painful or troubling emotions and playing up the pleasurable ones (love of their job, passion for science). This double move enables them to keep going, but in ways that systematically downplay worst-case scenarios and embody a kind of everyday denial favouring positive scenarios. By rationalizing emotions they systematically avoid worst-case scenarios, even where they are as statistically likely to occur as not.

5. Concluding implications

We have shown that emotion management works to:

- produce the subject of the scientist as objective and rational through strategies consistent with everyday denial (Norgaard, 2011),
- produce the collective of dedicated, inspired and passionate climate scientists,
- give form to the performance of an optimistic disposition which aligns with Erring on the Side of Least Drama (Brysse et al., 2013).

Our analysis of this group highlights that emotional management is an important part of dealing with climate change knowledge in both the professional and the domestic spheres. Our point is not to create more burdens and responsibilities for a group of dedicated professionals and citizens, nor are we suggesting that society should look to climate scientists for particular insights into how to behave. Rather this empirical study points to the complexity of dealing with climate change in a context of strong social and cultural pressure to be positive rather than negative. Recognizing how emotion works to align bodies and collectivities, how it is managed across domestic and professional realms and is important for sustaining a sense of self opens up thinking about climate change that goes beyond the rational and the scientific. It provokes a range of questions about the role of emotion in the climate change debate and enables us to be more explicit about how strategies of denial can play out (Opatow and Weiss, 2000). By attending to the emotional management strategies being employed by this group of scientists we hope that future research will build upon our findings and explore further possibilities. We might ask for example, how emotional management tools and strategies might be shared or common to diverse groups, or how painful or uncomfortable emotions might be managed to generate or produce effective action on climate change. Discussing how we feel about climate change will at the very least broaden our thinking and give us more options for effective action.

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